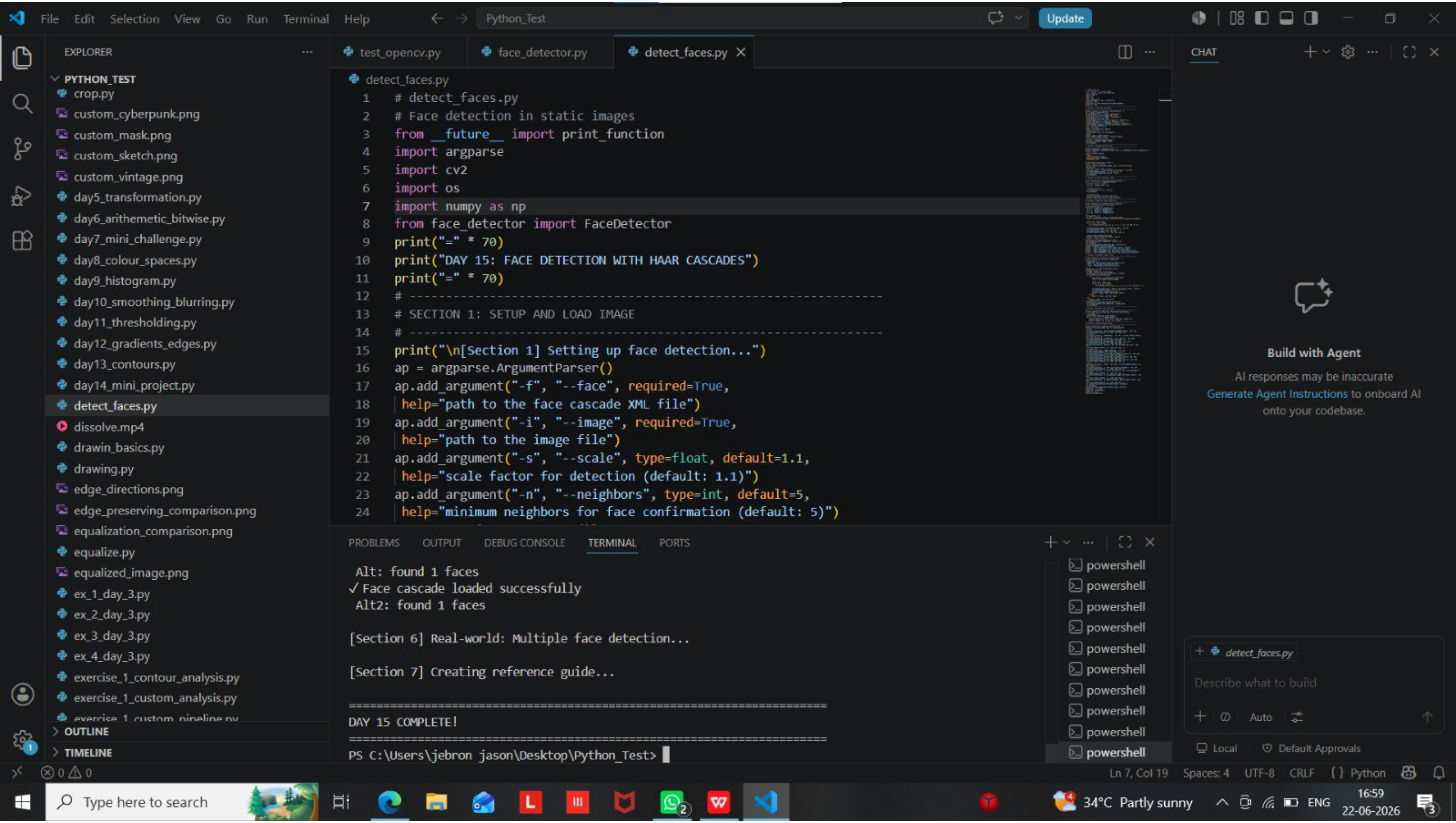
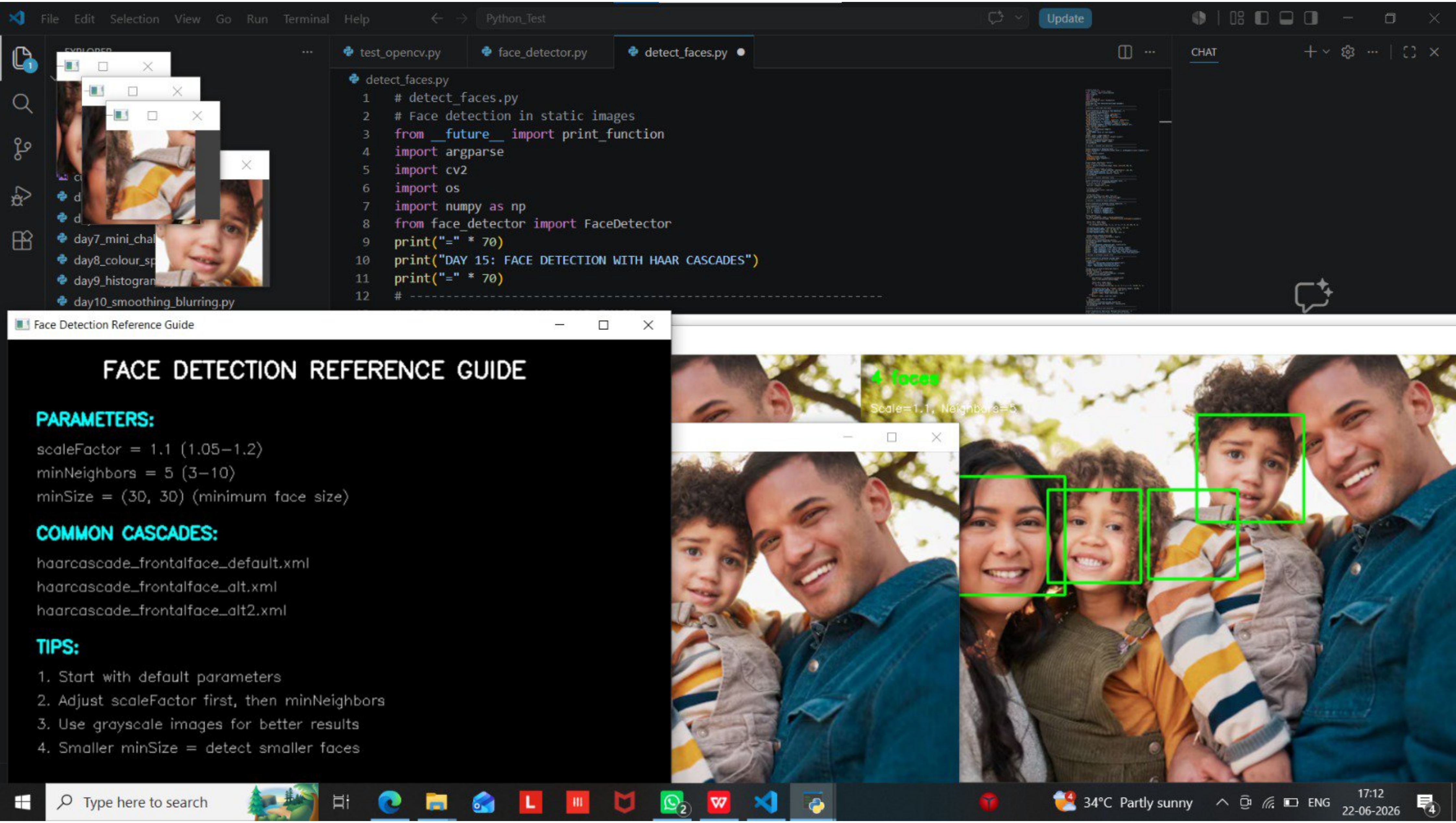
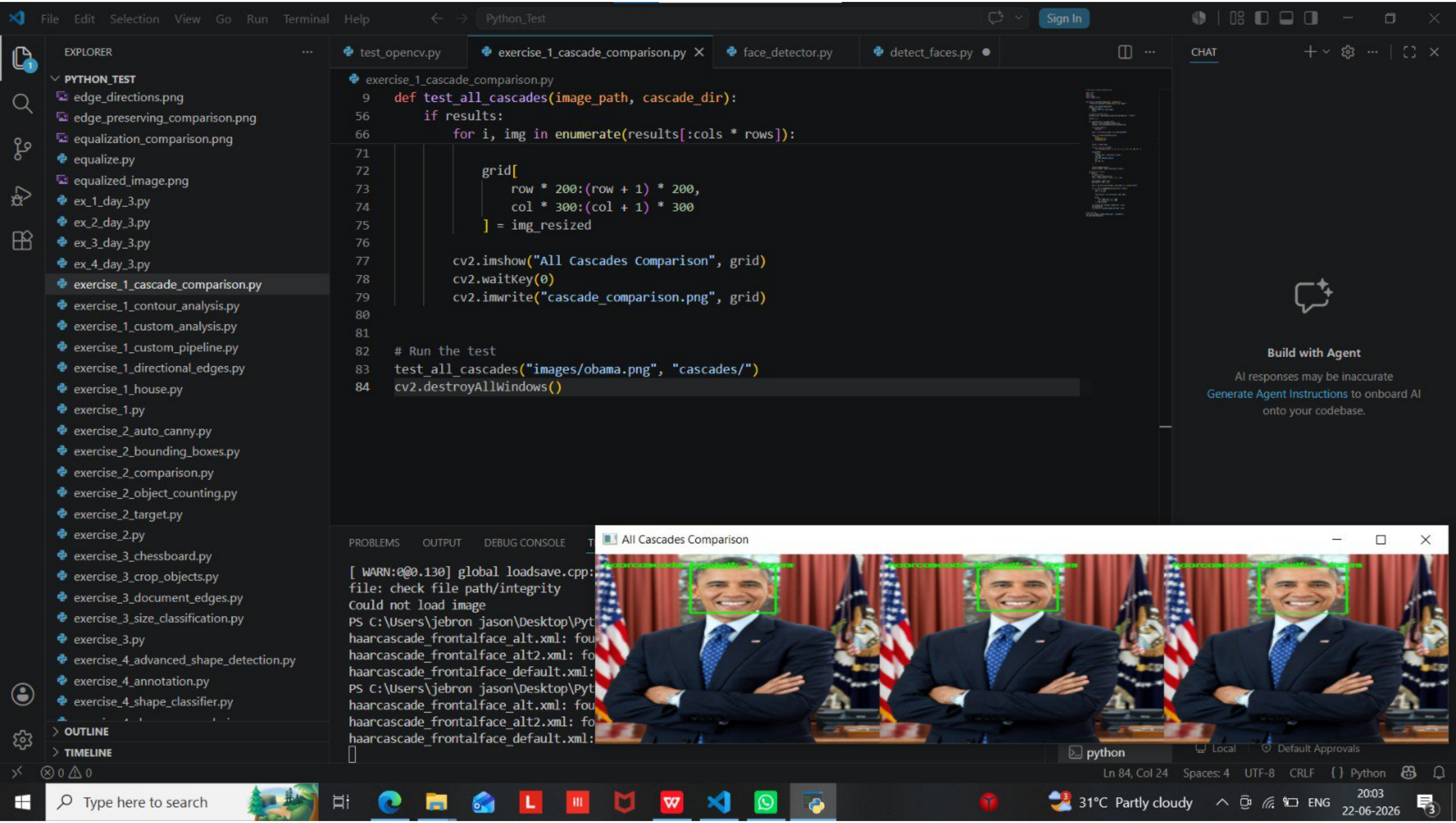
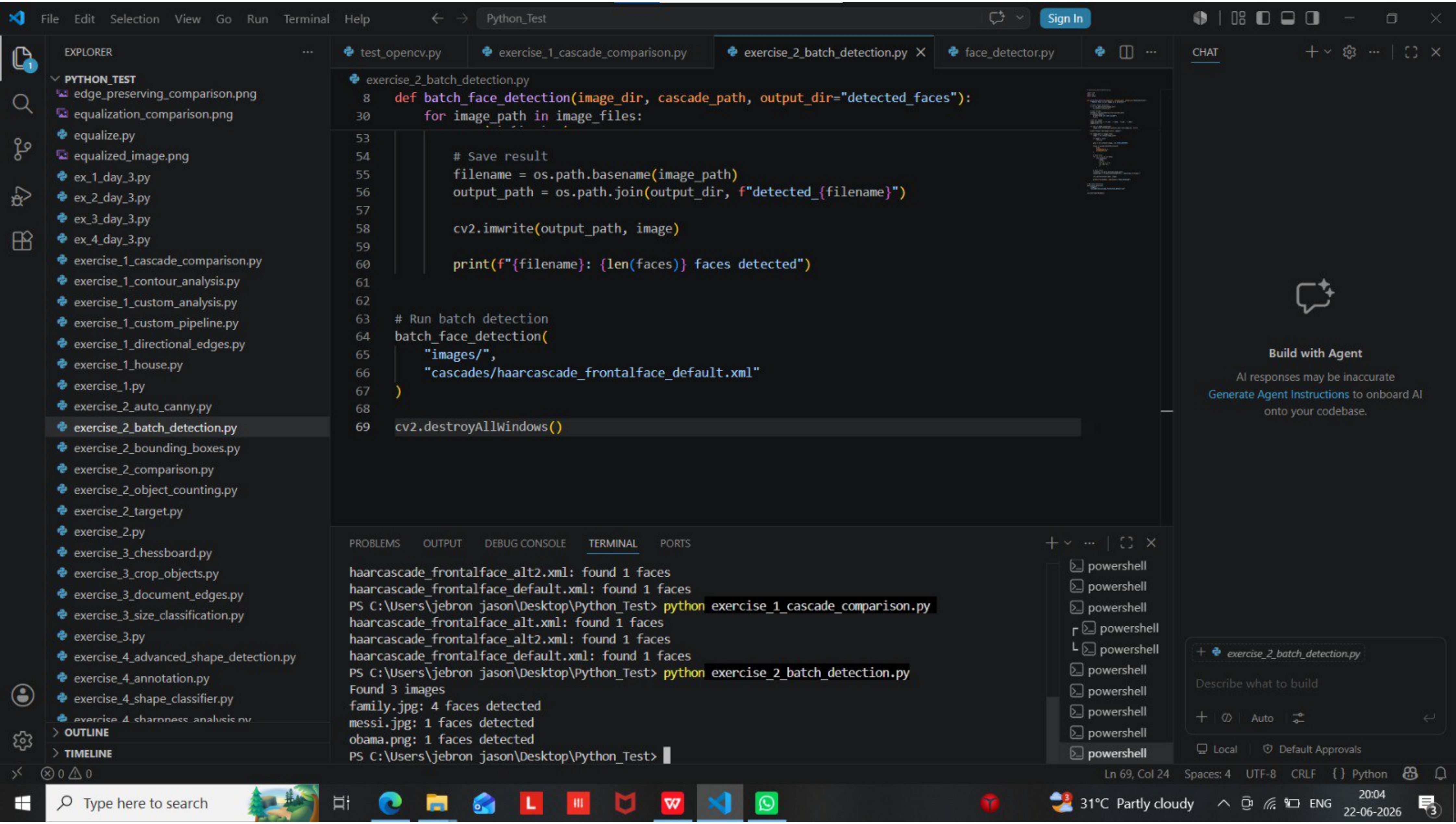


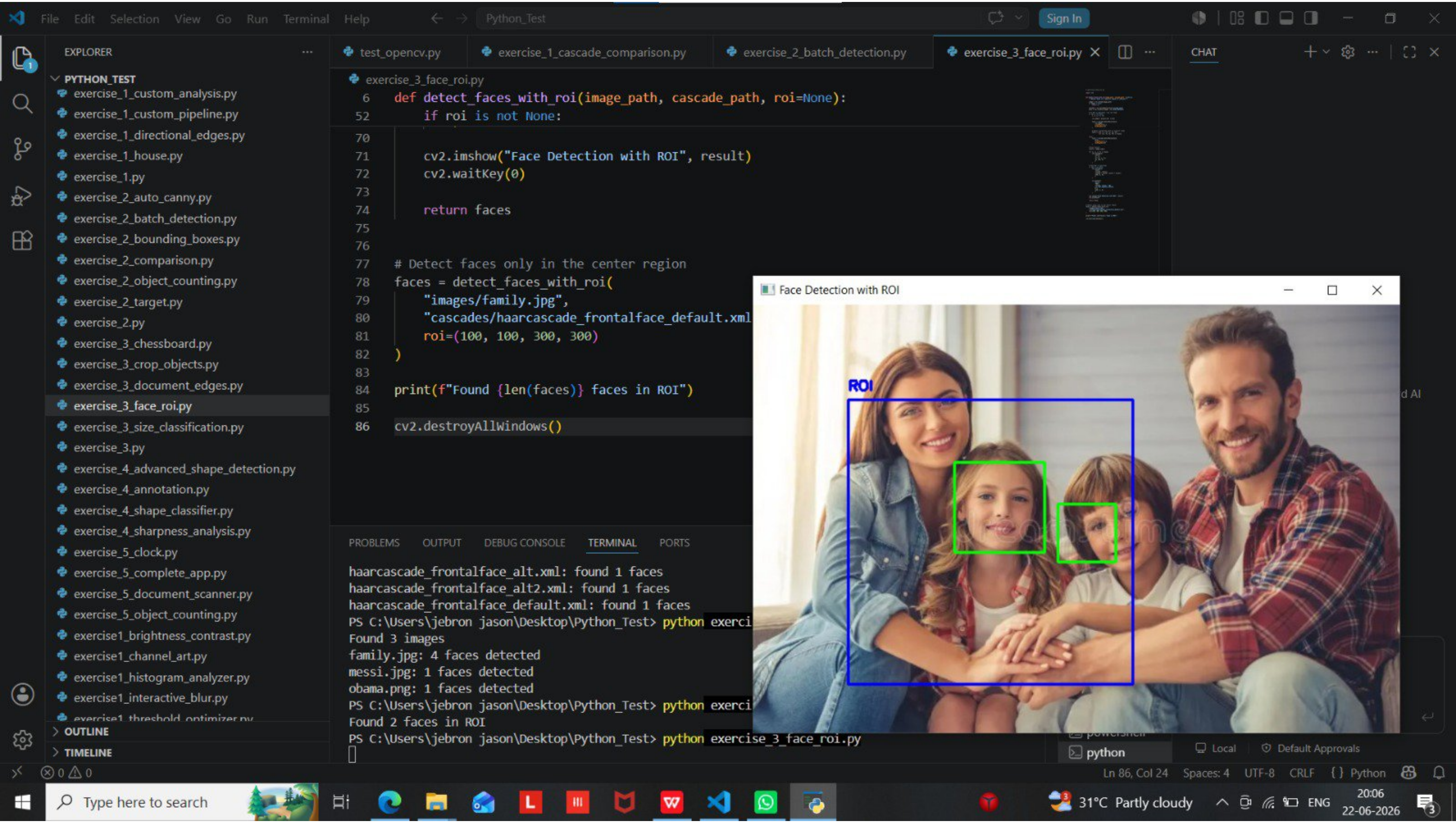
File Edit Selection View Go Run Terminal Help
Python_Test
Update
EXPLORER
PYTHON_TEST
crop.py
custom_cyberpunk.png
custom_mask.png
custom_sketch.png
custom_vintage.png
day5_transformation.py
day6_arithemetic_bitwise.py
day7_mini_challenge.py
day8_colour_spaces.py
day9_histogram.py
day10_smoothing_blurring.py
day11_thresholding.py
day12_gradients_edges.py
day13_contours.py
day14_mini_project.py
detect_faces.py
dissolve.mp4
drawin_basics.py
drawing.py
edge_directions.png
edge_preserving_comparison.png
equalization_comparison.png
equalize.py
equalized_image.png
ex_1_day_3.py
ex_2_day_3.py
ex_3_day_3.py
ex_4_day_3.py
exercise_1_contour_analysis.py
exercise_1_custom_analysis.py
OUTLINE
TIMELINE
test_opencv.py
face_detector.py
detect_faces.py
# detect_faces.py
# Face detection in static images
from __future__ import print_function
import argparse
import cv2
import os
import numpy as np
from face_detector import FaceDetector
print("=" * 70)
print("DAY 15: FACE DETECTION WITH HAAR CASCADES")
print("=" * 70)
# --
# SECTION 1: SETUP AND LOAD IMAGE
# --
print("\n[Section 1] Setting up face detection...")
ap = argparse.ArgumentParser()
ap.add_argument("-f", "--face", required=True,
help="path to the face cascade XML file")
ap.add_argument("-i", "--image", required=True,
help="path to the image file")
ap.add_argument("-s", "--scale", type=float, default=1.1,
help="scale factor for detection (default: 1.1)")
ap.add_argument("-n", "--neighbors", type=int, default=5,
help="minimum neighbors for face confirmation (default: 5)")
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
Alt: found 1 faces
✓ Face cascade loaded successfully
Alt2: found 1 faces
[Section 6] Real-world: Multiple face detection...
[Section 7] Creating reference guide...
==
DAY 15 COMPLETE!
==
PS C:\Users\jebron jason\Desktop\Python_Test>
powershell
CHAT
Build with Agent
AI responses may be inaccurate
Generate Agent Instructions to onboard AI onto your codebase.
detect_faces.py
Describe what to build
Auto
Local
Default Approvals
Ln 7, Col 19 Spaces: 4 UTF-8 CRLF { } Python
Type here to search
34°C Partly sunny
ENG
16:59
22-06-2026

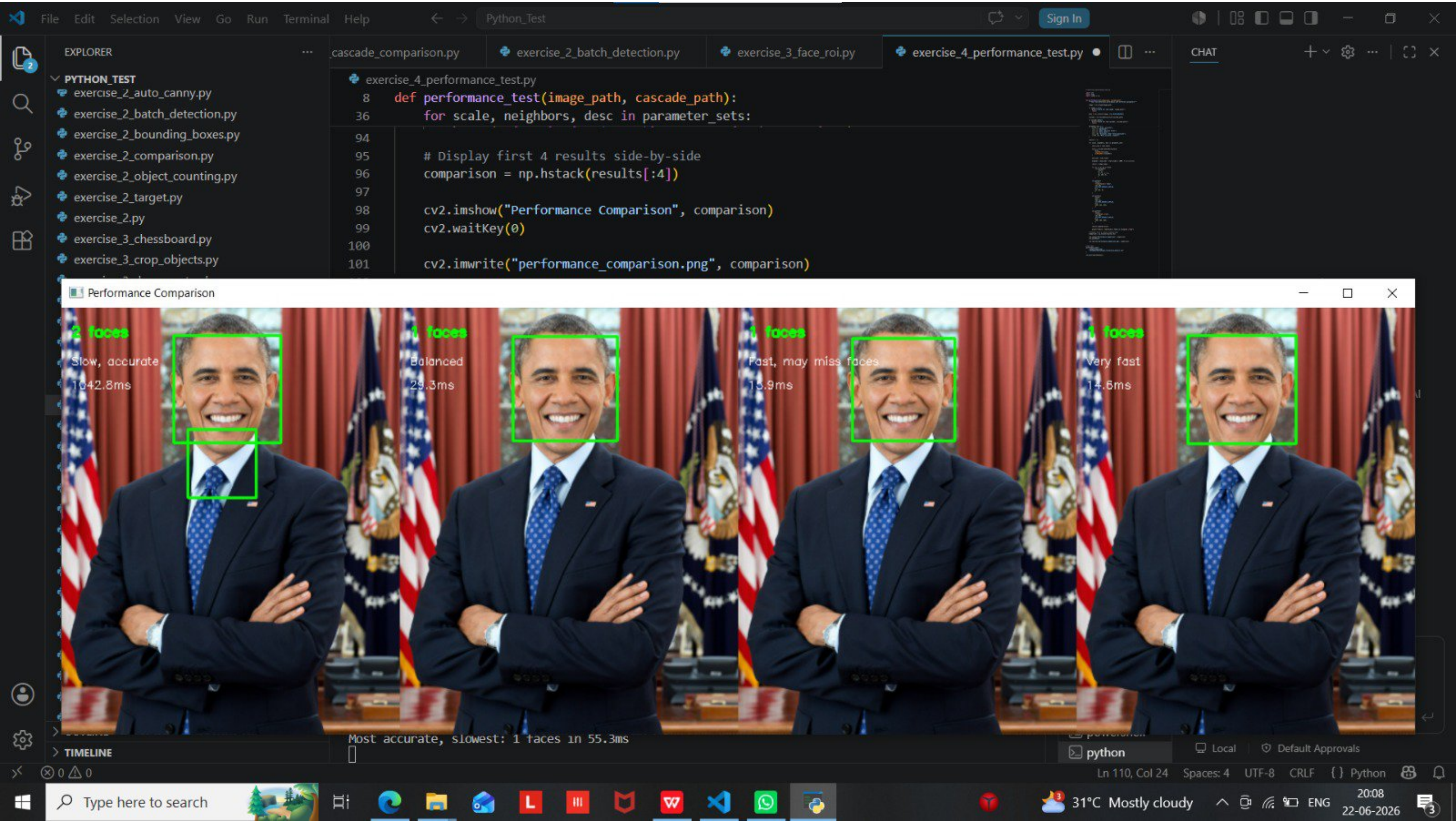


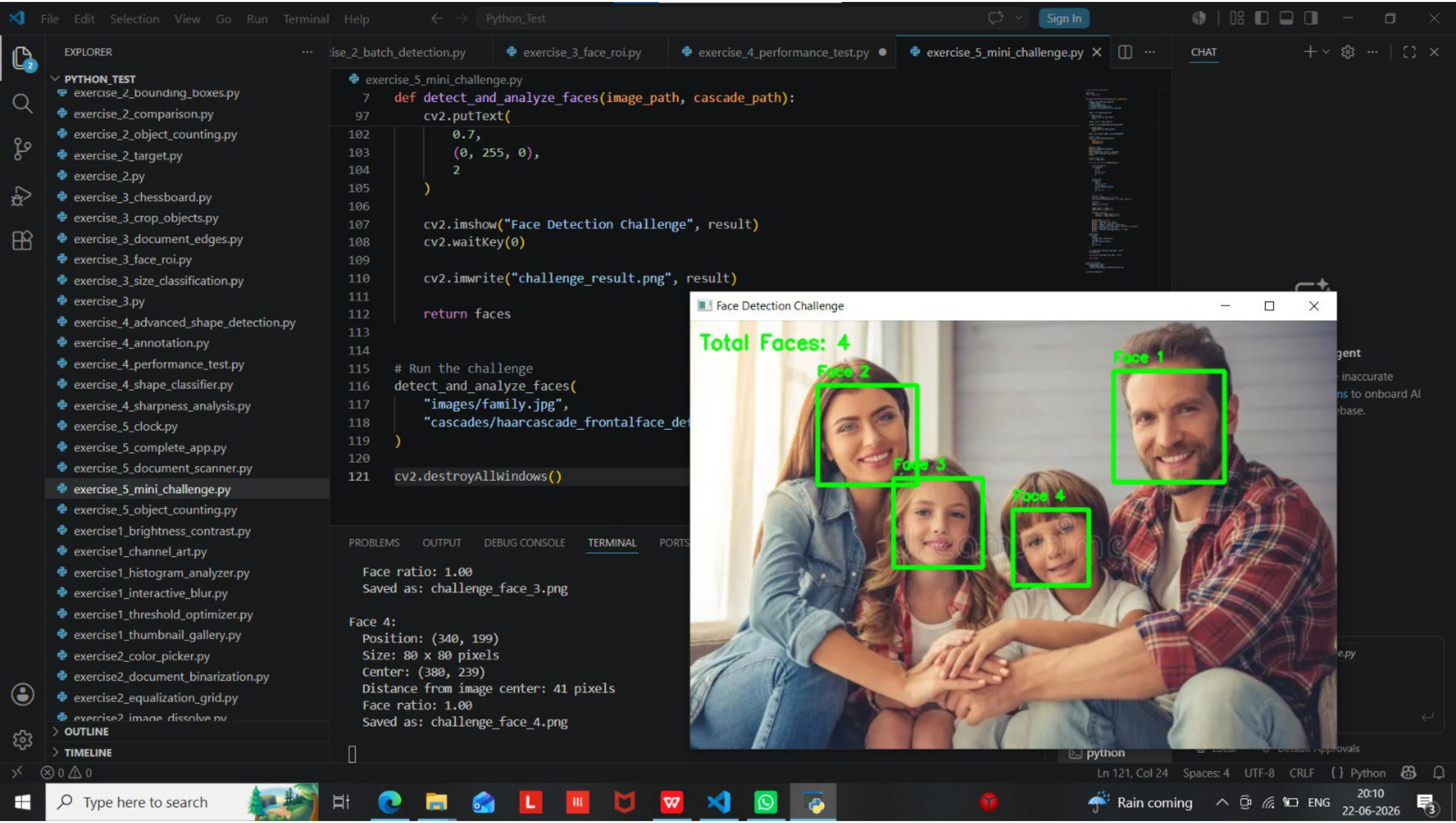
File Edit Selection View Go Run Terminal Help
Python_Test
Update
test_opencv.py
face_detector.py
detect_faces.py
CHAT
day7_mini_chal
day8_colour_sp
day9_histogran
day10_smoothing_blurring.py
detect_faces.py
# detect_faces.py
# Face detection in static images
from __future__ import print_function
import argparse
import cv2
import os
import numpy as np
from face_detector import FaceDetector
print("=" * 70)
print("DAY 15: FACE DETECTION WITH HAAR CASCADES")
print("=" * 70)
# --
Face Detection Reference Guide
FACE DETECTION REFERENCE GUIDE
PARAMETERS:
scaleFactor = 1.1 (1.05-1.2)
minNeighbors = 5 (3-10)
minSize = (30, 30) (minimum face size)
COMMON CASCADES:
haarcascade_frontalface_default.xml
haarcascade_frontalface_alt.xml
haarcascade_frontalface_alt2.xml
TIPS:
1. Start with default parameters
2. Adjust scaleFactor first, then minNeighbors
3. Use grayscale images for better results
4. Smaller minSize = detect smaller faces
4 faces
Type here to search
34°C Partly sunny
ENG
17:12
22-06-2026


File Edit Selection View Go Run Terminal Help
Python_Test
Sign In
EXPLORER
PYTHON_TEST
edge_directions.png
edge_preserving_comparison.png
equalization_comparison.png
equalize.py
equalized_image.png
ex_1_day_3.py
ex_2_day_3.py
ex_3_day_3.py
ex_4_day_3.py
exercise_1_cascade_comparison.py
exercise_1_contour_analysis.py
exercise_1_custom_analysis.py
exercise_1_custom_pipeline.py
exercise_1_directional_edges.py
exercise_1_house.py
exercise_1.py
exercise_2_auto_canny.py
exercise_2_bounding_boxes.py
exercise_2_comparison.py
exercise_2_object_counting.py
exercise_2_target.py
exercise_2.py
exercise_3_chessboard.py
exercise_3_crop_objects.py
exercise_3_document_edges.py
exercise_3_size_classification.py
exercise_3.py
exercise_4_advanced_shape_detection.py
exercise_4_annotation.py
exercise_4_shape_classifier.py
OUTLINE
TIMELINE
test_opencv.py
exercise_1_cascade_comparison.py
face_detector.py
detect_faces.py
def test_all_cascades(image_path, cascade_dir):
    if results:
        for i, img in enumerate(results[:cols * rows]):
            grid[
                row * 200:(row + 1) * 200,
                col * 300:(col + 1) * 300
            ] = img_resized
        cv2.imshow("All Cascades Comparison", grid)
        cv2.waitKey(0)
        cv2.imwrite("cascade_comparison.png", grid)
# Run the test
test_all_cascades("images/obama.png", "cascades/")
cv2.destroyAllWindows()
PROBLEMS OUTPUT DEBUG CONSOLE
[WARN:0@0.130] global loadsave.cpp:
file: check file path/integrity
Could not load image
PS C:\Users\jebron jason\Desktop\Pyt
haarcascade_frontalface_alt.xml: fou
haarcascade_frontalface_alt2.xml: fo
haarcascade_frontalface_default.xml:
PS C:\Users\jebron jason\Desktop\Pyt
haarcascade_frontalface_alt.xml: fou
haarcascade_frontalface_alt2.xml: fo
haarcascade_frontalface_default.xml:
All Cascades Comparison
python
Local
Default Approvals
CHAT
Build with Agent
AI responses may be inaccurate
Generate Agent Instructions to onboard AI onto your codebase.
Ln 84, Col 24 Spaces: 4 UTF-8 CRLF { } Python
Type here to search
31°C Partly cloudy
ENG
20:03
22-06-2026


File Edit Selection View Go Run Terminal Help
Python_Test
Sign In
EXPLORER
PYTHON_TEST
edge_preserving_comparison.png
equalization_comparison.png
equalize.py
equalized_image.png
ex_1_day_3.py
ex_2_day_3.py
ex_3_day_3.py
ex_4_day_3.py
exercise_1_cascade_comparison.py
exercise_1_contour_analysis.py
exercise_1_custom_analysis.py
exercise_1_custom_pipeline.py
exercise_1_directional_edges.py
exercise_1_house.py
exercise_1.py
exercise_2_auto_canny.py
exercise_2_batch_detection.py
exercise_2_bounding_boxes.py
exercise_2_comparison.py
exercise_2_object_counting.py
exercise_2_target.py
exercise_2.py
exercise_3_chessboard.py
exercise_3_crop_objects.py
exercise_3_document_edges.py
exercise_3_size_classification.py
exercise_3.py
exercise_4_advanced_shape_detection.py
exercise_4_annotation.py
exercise_4_shape_classifier.py
OUTLINE
TIMELINE
test_opencv.py
exercise_1_cascade_comparison.py
exercise_2_batch_detection.py
face_detector.py
def batch_face_detection(image_dir, cascade_path, output_dir="detected_faces"):
for image_path in image_files:
# Save result
filename = os.path.basename(image_path)
output_path = os.path.join(output_dir, f"detected_{filename}")
cv2.imwrite(output_path, image)
print(f"{filename}: {len(faces)} faces detected")
# Run batch detection
batch_face_detection(
"images/",
"cascades/haarcascade_frontalface_default.xml"
)
cv2.destroyAllWindows()
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
haarcascade_frontalface_alt2.xml: found 1 faces
haarcascade_frontalface_default.xml: found 1 faces
PS C:\Users\jebron jason\Desktop\Python_Test> python exercise_1_cascade_comparison.py
haarcascade_frontalface_alt.xml: found 1 faces
haarcascade_frontalface_alt2.xml: found 1 faces
haarcascade_frontalface_default.xml: found 1 faces
PS C:\Users\jebron jason\Desktop\Python_Test> python exercise_2_batch_detection.py
Found 3 images
family.jpg: 4 faces detected
messi.jpg: 1 faces detected
obama.png: 1 faces detected
PS C:\Users\jebron jason\Desktop\Python_Test>
powershell
CHAT
Build with Agent
AI responses may be inaccurate
Generate Agent Instructions to onboard AI onto your codebase.
Describe what to build
Auto
Local
Default Approvals
Ln 69, Col 24 Spaces: 4 UTF-8 CRLF { } Python
Type here to search
31°C Partly cloudy
ENG
20:04
22-06-2026


File Edit Selection View Go Run Terminal Help
Python_Test
Sign In
EXPLORER
PYTHON_TEST
exercise_1_custom_analysis.py
exercise_1_custom_pipeline.py
exercise_1_directional_edges.py
exercise_1_house.py
exercise_1.py
exercise_2_auto_canny.py
exercise_2_batch_detection.py
exercise_2_bounding_boxes.py
exercise_2_comparison.py
exercise_2_object_counting.py
exercise_2_target.py
exercise_2.py
exercise_3_chessboard.py
exercise_3_crop_objects.py
exercise_3_document_edges.py
exercise_3_face_roi.py
exercise_3_size_classification.py
exercise_3.py
exercise_4_advanced_shape_detection.py
exercise_4_annotation.py
exercise_4_shape_classifier.py
exercise_4_sharpness_analysis.py
exercise_5_clock.py
exercise_5_complete_app.py
exercise_5_document_scanner.py
exercise_5_object_counting.py
exercise1_brightness_contrast.py
exercise1_channel_art.py
exercise1_histogram_analyzer.py
exercise1_interactive_blur.py
OUTLINE
TIMELINE
test_opencv.py
exercise_1_cascade_comparison.py
exercise_2_batch_detection.py
exercise_3_face_roi.py
CHAT
def detect_faces_with_roi(image_path, cascade_path, roi=None):
if roi is not None:
cv2.imshow("Face Detection with ROI", result)
cv2.waitKey(0)
return faces
# Detect faces only in the center region
faces = detect_faces_with_roi(
"images/family.jpg",
"cascades/haarcascade_frontalface_default.xml
roi=(100, 100, 300, 300)
)
print(f"Found {len(faces)} faces in ROI")
cv2.destroyAllWindows()
Face Detection with ROI
ROI
PROBLEMS
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
haarcascade_frontalface_alt.xml: found 1 faces
haarcascade_frontalface_alt2.xml: found 1 faces
haarcascade_frontalface_default.xml: found 1 faces
PS C:\Users\jebron jason\Desktop\Python_Test> python exerci
Found 3 images
family.jpg: 4 faces detected
messi.jpg: 1 faces detected
obama.png: 1 faces detected
PS C:\Users\jebron jason\Desktop\Python_Test> python exerci
Found 2 faces in ROI
PS C:\Users\jebron jason\Desktop\Python_Test> python exercise_3_face_roi.py
python
Local
Default Approvals
Ln 86, Col 24
Spaces: 4
UTF-8
CRLF
{ } Python
Type here to search
31°C Partly cloudy
ENG
20:06
22-06-2026


File Edit Selection View Go Run Terminal Help
Python_Test
Sign In
EXPLORER
PYTHON_TEST
exercise_2_auto_canny.py
exercise_2_batch_detection.py
exercise_2_bounding_boxes.py
exercise_2_comparison.py
exercise_2_object_counting.py
exercise_2_target.py
exercise_2.py
exercise_3_chessboard.py
exercise_3_crop_objects.py
cascade_comparison.py
exercise_2_batch_detection.py
exercise_3_face_roi.py
exercise_4_performance_test.py
CHAT
def performance_test(image_path, cascade_path):
for scale, neighbors, desc in parameter_sets:
# Display first 4 results side-by-side
comparison = np.hstack(results[:4])
cv2.imshow("Performance Comparison", comparison)
cv2.waitKey(0)
cv2.imwrite("performance_comparison.png", comparison)
Performance Comparison
2 faces
Slow, accurate
1 faces
Balanced
1 faces
Fast, may miss faces
1 faces
Very fast
Most accurate, slowest: 1 faces in 55.3ms
TIMELINE
python
Local
Default Approvals
Ln 110, Col 24
Spaces: 4
UTF-8
CRLF
Python
Type here to search
31°C Mostly cloudy
ENG
20:08
22-06-2026


File Edit Selection View Go Run Terminal Help
Python_Test
Sign In
EXPLORER
PYTHON_TEST
exercise_2_bounding_boxes.py
exercise_2_comparison.py
exercise_2_object_counting.py
exercise_2_target.py
exercise_2.py
exercise_3_chessboard.py
exercise_3_crop_objects.py
exercise_3_document_edges.py
exercise_3_face_roi.py
exercise_3_size_classification.py
exercise_3.py
exercise_4_advanced_shape_detection.py
exercise_4_annotation.py
exercise_4_performance_test.py
exercise_4_shape_classifier.py
exercise_4_sharpness_analysis.py
exercise_5_clock.py
exercise_5_complete_app.py
exercise_5_document_scanner.py
exercise_5_mini_challenge.py
exercise_5_object_counting.py
exercise1_brightness_contrast.py
exercise1_channel_art.py
exercise1_histogram_analyzer.py
exercise1_interactive_blur.py
exercise1_threshold_optimizer.py
exercise1_thumbnail_gallery.py
exercise2_color_picker.py
exercise2_document_binarization.py
exercise2_equalization_grid.py
OUTLINE
TIMELINE
ise_2_batch_detection.py
exercise_3_face_roi.py
exercise_4_performance_test.py
exercise_5_mini_challenge.py
CHAT
7 def detect_and_analyze_faces(image_path, cascade_path):
97 cv2.putText(
102 0.7,
103 (0, 255, 0),
104 2
105)
107 cv2.imshow("Face Detection Challenge", result)
108 cv2.waitKey(0)
110 cv2.imwrite("challenge_result.png", result)
112 return faces
115 # Run the challenge
116 detect_and_analyze_faces(
117 "images/family.jpg",
118 "cascades/haarcascade_frontalface_def
119)
121 cv2.destroyAllWindows()
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
Face ratio: 1.00
Saved as: challenge_face_3.png
Face 4:
Position: (340, 199)
Size: 80 x 80 pixels
Center: (380, 239)
Distance from image center: 41 pixels
Face ratio: 1.00
Saved as: challenge_face_4.png
Face Detection Challenge
Total Faces: 4
Face 1
Face 2
Face 3
Face 4
Ln 121, Col 24 Spaces: 4 UTF-8 CRLF { } Python
Type here to search
Rain coming
ENG
20:10
22-06-2026